



Team Details

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Problem Statement Addressed



AI-Based Restoration of Degraded Images for Semiconductor Inspection | KLA, Track 1

DESCRIPTION / DETAILS

Inspection images from a semiconductor line arrive degraded three ways at once: speckle noise (multiplicative, and it pushes pixel values outside the true intensity range), additive Gaussian noise (soft, hazy edges), and 2x spatial downsampling (fine detail genuinely lost, not merely hidden). Any combination and strength may be present in a single image.

Why it matters: *these degradations hide the very features an inspection system exists to detect. Blur the noise away and real structure is destroyed; sharpen too hard and structure appears that was never there. Both failure modes translate directly into missed defects or false defect calls on a production line.*

The task: *learn one transformation from paired (degraded, clean) examples that reverses all three degradations simultaneously, generalises to structure types never seen in training, and runs fast enough to keep up with inline inspection throughput.*



Idea Description



A single compact CNN that restores all three degradations in one forward pass, and adapts its own processing to how badly each individual image is degraded.

KEY CONCEPT & APPROACH

We treat this as one joint problem rather than three sequential ones. A NAFNet-lite encoder-decoder (LayerNorm + depthwise conv + SimpleGate + simplified channel attention) does denoising, deblurring and 2x super-resolution together in a single pass, ending in a PixelShuffle upsample head.

Staging them separately would let each stage's errors compound into the next; solving them jointly avoids that and is far faster at inference.

On top of that, a small degradation estimator reads each input and emits a 32-d vector describing how degraded that specific image is. A FiLM layer turns that vector into a per-channel scale and shift applied at the bottleneck, so the network adapts per image instead of applying one fixed strategy.

SOLUTION OVERVIEW

Speckle noise -> Charbonnier loss is robust to the outlier pixels speckle creates, and inputs are never clipped on load, so out-of-range values are preserved.

Gaussian blur -> a Sobel edge loss explicitly penalises smeared edges, directly targeting the 'do not blur the image to remove noise' constraint.

Downsampling -> a learned PixelShuffle head reconstructs the 2x detail rather than interpolating it. The base model uses no adversarial loss at all; a low-weight GAN term (0.015) is introduced only in a later fine-tune stage, deliberately capped low so it cannot dominate — unconstrained GAN-style training invents plausible texture, which is exactly the ringing and artificial patterning the problem statement warns against.



Proposed Solution

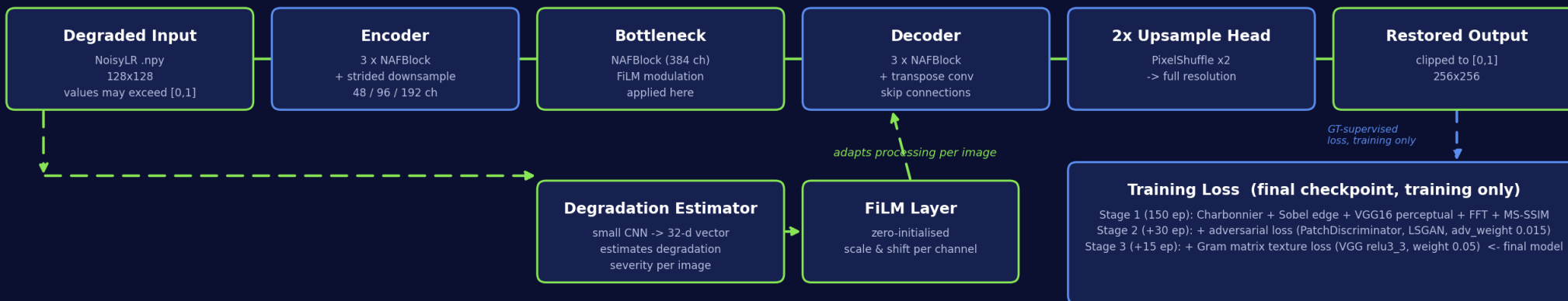
Final submitted pipeline (NAFNet+Gram). The dashed green path is our degradation-aware FiLM conditioning; the dashed blue path is the training-only loss supervision, built up over three fine-tuning stages.

NAFNet + Gram — Final Restoration Pipeline

Single forward pass: joint denoise + 2x super-resolution, conditioned on per-image degradation severity

— adaptive conditioning path (our novelty)

— supervision signal, training only — not used at inference



Checkpoint lineage: best.pth (150 ep, val_loss 0.120) -> 180-epoch GAN fine-tune (increased_quality.pth) -> Gram fine-tune, 15 ep = nafnet+gram_best.pth (FINAL, validated 30/30 vs. previous best)



Innovation and Uniqueness



Most submissions will train one fixed restoration network. Ours estimates degradation severity per image and adapts to it — and we independently validated that claim, not just asserted it.

KEY INNOVATION

Degradation-aware FiLM conditioning. A small estimator network reads each degraded input and predicts how much damage it carries; a FiLM layer converts that into a per-channel scale and shift at the bottleneck. The network therefore processes a lightly-degraded image differently from a heavily-degraded one, instead of averaging one strategy over both.

The FiLM layer is zero-initialised, so training starts exactly equivalent to the plain baseline and can only deviate if doing so measurably lowers the loss. Our first attempt used default random init and lost to the baseline on every metric; diagnosing that and fixing the initialisation is what turned the idea into a real gain.

COMPETITIVE ADVANTAGE

Validated with real evidence, not just a diagram. A dedicated post-training diagnostic on our final checkpoint shows FiLM's learned weights are clearly non-zero (active, not a no-op), and its response magnitude correlates with independently-measured noise severity across 45 images spanning the full dataset.

Statistically confirmed, not a coincidence. A shuffle test (2,000 random re-pairings) shows this correlation could not have happened by chance ($p < 0.0001$). Direct ablation further shows the suppression is necessary to prevent artifacts at high noise, with a narrower effect at low noise — an honest, evidenced result.



FiLM Novelty — Validated



Our mentor asked us to define this novelty properly. So we didn't just describe the mechanism — we ran five direct tests on the final checkpoint and report the numbers, including where the effect is limited.

DIAGNOSTIC TEST

FiLM is alive, not a no-op. Its learned scale and shift weight norms (15.64 / 2.80 / 2.89 / 0.14) are all clearly non-zero — it genuinely deviated from its zero-init start.

It tracks real degradation. Across 45 images sampled to span the full dataset (severity 0.021 to 0.381), FiLM's response magnitude correlates with measured noise severity: $r = 0.576$ — not a straight line, but real.

It's not chance. A shuffle test (2,000 random re-pairings of the same 45 values) gives $p < 0.0001$: 0 of 2,000 random pairings matched or beat the real correlation.

ABLATION TEST

Ablation-tested, not just theorised. We manually override FiLM's scale on real images and re-ran the model to see what its heavy suppression (~2-7% passthrough) is for.

It's necessary at high noise. Removing it causes visible grid/checkerboard artifacts on our highest-noise test image — suppression is protecting the output there.

It's not the cause of our blur issue at low noise. On a low-noise image, reducing suppression added no visible detail — pointing back to loss-function design instead.



Results

Head-to-head architecture comparison

Same data, same held-out crops — final RRDB checkpoint vs. final NAFNet+Gram checkpoint

PSNR / SSIM: higher is better | LPIPS: lower is better

Model	PSNR (dB)	SSIM	LPIPS
RRDB (115 ep)	22.802	0.544	0.190 (best)
NAFNet + Gram (final)	23.646 (best)	0.597 (best)	0.246

Honest trade-off: NAFNet+Gram wins decisively on PSNR and SSIM. RRDB scores

better (lower) LPIPS on this comparison — we report it rather than omit it.

Corroborating evidence: in a separate, broader 40-image comparison, NAFNet

won 29/40 images outright vs. RRDB's 8/40 (3 ties) — architecture choice

confirmed across two independent evaluations, not a single lucky run.

30 / 30

images where the Gram fine-tune beat the pre-Gram baseline checkpoint, head-to-head

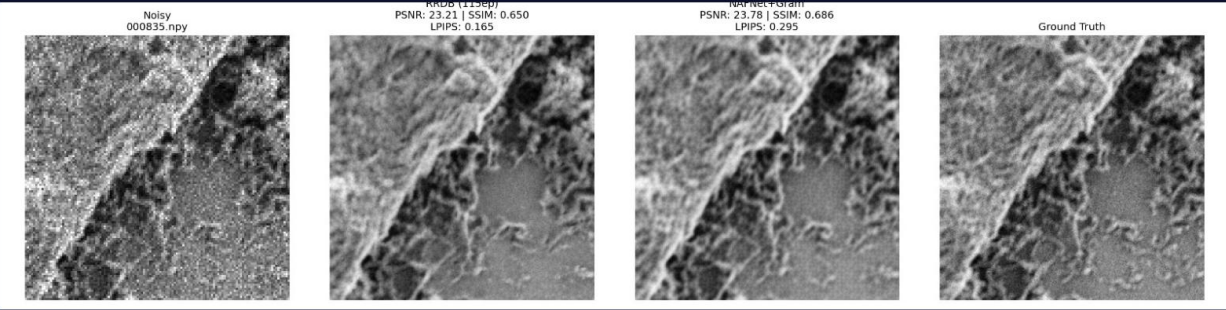
23.09 → 23.56 dB

average PSNR, pre-Gram baseline vs. final NAFNet+Gram checkpoint

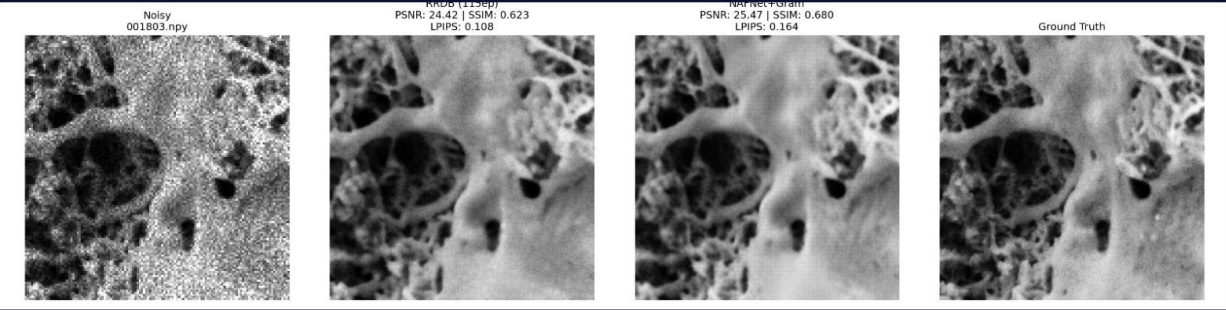
29 / 40

images where NAFNet beat RRDB outright in the broader architecture comparison

Visual results — Noisy | RRDB (115 ep) | NAFNet+Gram (final) | Ground Truth



000835.npy — fine crack/filament structure: NAFNet+Gram keeps sharper boundaries



001803.npy — dense pore structure: NAFNet+Gram best matches Ground Truth edges



Technology & Feasibility/Methodology Used



PyTorch throughout. Trained on a single free-tier Colab T4; the whole five-model programme fits in a few GPU-hours, so every result here is reproducible on hardware any student can access.

IMPLEMENTATION STRATEGY

Training: PyTorch, Adam (lr 1e-3), cosine annealing, batch 16, 3-stage pipeline (150+30+15=195 epochs total), synthetic + elastic augmentation, seed 42 for a reproducible split.

Losses: Charbonnier + Sobel edge + VGG16 perceptual + FFT + MS-SSIM (Stage 1), + low-weight adversarial (Stage 2), + Gram matrix texture loss (Stage 3, final).

Model: 2.73 M params, kept small because throughput is scored.

Inference: run.py takes --input_dir / --output_dir, batches same-shape images, auto-detects GPU, clips output to [0,1].

Metrics: PSNR/SSIM via scikit-image, LPIPS via the standard lpips package - reference implementations, not hand-rolled.



PyTorch . NumPy . scikit-image



Colab T4 train . H100 eval



Git . VS Code . Colab



GitHub & Video Link



GitHub Repository

<https://github.com/dharshanrajurd/TECHNOX-NoiseTrue-Adaptive-Phase-2-Semicon>



Prototype / Simulation Video

<https://youtu.be/M1pKS-XS6pU>



Research and References



Research Background & Methodology

Every architectural and loss choice below is taken from published restoration literature rather than invented for this submission.

NAFNet supplies the backbone block. SimpleGate replaces nonlinear activations, and simplified channel attention replaces heavier attention, giving a better quality-per-FLOP trade-off than a plain U-Net. FiLM supplies the conditioning mechanism - a feature-wise affine modulation driven by a side network.

Charbonnier is a differentiable robust L1 variant, appropriate here because speckle produces genuine outlier pixels. The VGG perceptual loss follows the standard feature-space formulation, and LPIPS (our evaluation metric) is built on the same principle, which is why optimising the former moves the latter.



References & Citations

Papers, libraries and pretrained weights used. All licences permit competition use.

Chen, L. et al. (2022). Simple Baselines for Image Restoration (NAFNet). ECCV 2022. arxiv.org/abs/2204.04676

Perez, E. et al. (2018). FiLM: Visual Reasoning with a General Conditioning Layer. AAAI 2018. arxiv.org/abs/1709.07871

Zhang, R. et al. (2018). The Unreasonable Effectiveness of Deep Features as a Perceptual Metric (LPIPS). CVPR 2018. arxiv.org/abs/1801.03924

Simonyan, K. & Zisserman, A. (2015). Very Deep Convolutional Networks for Large-Scale Image Recognition (VGG16). ICLR 2015. arxiv.org/abs/1409.1556
Dataset: KLA paired GT / NoisyLR training set, SEMICON India Hackathon 2026. Pretrained VGG16 weights via torchvision (BSD-3); lpips package (BSD-2)